MIL-E-1/1342B
6 December 1971
SUPERSEDING
MIL-E-1/1342A
1 December 1961

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 1EP1

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Miniature, 1-inch tube, electrostatic deflection and focus

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	ec1	ed	Eb1	Eb2	Ehk	Rg1	Rd Note 1	Alt
Unit:	V	Vdc	v	v	Vdc	Vdc	Vdc	Meg	Meg	ft
Maximum:	6.9	0	2	500	1,200	1,750	+125	1.5	2.0	20,000
Minimum:	5.7	-200	---	---	---	500	-125	---	---	---
TEST CONDITIONS:	6.3	Vary	---	---	Focus	500	---	---	---	---

GENERAL:

Qualification - Required

(B) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
1002	Barometric pressure, reduced	---	Ef = 6.3 V; Ec1 = -200 Vdc; Eb1 = 1,200 Vdc; Eb2 = 1,700 Vdc; ED1 = ED2 = ED3 = ED4 = 2,250 Vdc; Ehk = 125 Vdc; Rg1 = 1.5 Meg; Rd = 2.0 Meg	---	350	---	mmHg
	Quality conformance inspection, part 1						
5201	Voltage breakdown (electrostatic types)	3	Eb2 = 1,750 Vdc; Eb1 = 1,200 Vdc; Ec1 = -200 Vdc	---	---	---	---
5206	Gas "cross"	2	Ec1 = 0; pattern size: 2 x 2 cm	---	---	---	---
5106	Bulb, screen, and faceplate quality	---		---	---	---	---
5241	Grid-cutoff voltage	---		Ec1	-7	-21	Vdc
5231	Spot position	4		---	---	2.5	mm
(B) 5251	Grid No. 1 leakage current	---		Ic1	---	5	µAdc
(B) 5251	Anode No. 2 leakage current	---		Ib2	---	15	µAdc
	Quality conformance inspection, part 2						
1301	Heater current	---		If	540	660	mA
5101	Base alinement (electrostatic types)	---	3D4: midpoint between pins No. 9 and No. 10 at base	---	---	---	---
5101	Angle between traces	---		---	---	---	---
5221	Light output	---	Ec1 = 0; pattern size: 2 x 2 cm, scan frequency = 60 x 2,940 Hz (49 lines)	---	4.5	---	ftL
5246	Focusing voltage at cutoff	---	Pattern: spot	Ec3	---	150	Vdc
5246	Focusing voltage (zero bias)	---	Pattern size: 2 x 2 cm; scan freq = 60 x 2,940 Hz (49 lines)	Ec3	50	---	Vdc
5248	Deflection factor	---	1D2	DF	105	155	Vdc/in.
5248	Deflection factor	---	3D4	DF	120	175	Vdc/in.

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued						
5251	Heater-cathode leakage current	---		Ihk	---	30	μAdc
(B) 5226	Line width A (electrostatic deflection)	---	Ik = 10 μA	Width	---	0.51	mm
(B) 5226	Line width B (electrostatic deflection)	---	Ik = 10 μA; 0.8 cm from center	Width	---	0.53	mm
(B) 1111	Base pin solder depth	---		---	---	---	---
(B) 1105	Permance of marking	---		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	---	Group C; Ec1/Ik = 6 μA; Eb2 = 1,750 Vdc; t = 500 hours (min)	---	---	---	---
---	Life-test end points:						
5226	Line width A	---	Ik = 10 μA	Width	---	0.51	mm
5226	Line width B	---	Ik = 10 μA; 0.8 cm from center	Width	---	0.53	mm

NOTES:

1. Rd is resistance of deflection plate current.
2. This test to be performed at the conclusion of the holding period.
3. Reject for continuous arcing.
4. The true beam position shall fall within a circle of the indicated radius concentric with tube face.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES

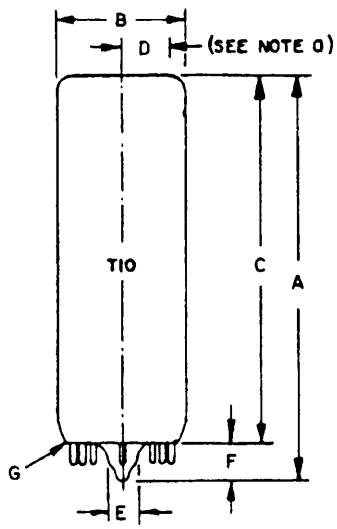
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

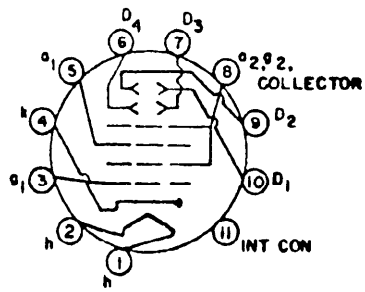
Agent: DSA - ES

(Project 5960-2559)

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ORIENTATION OF DEFLECTION PLATES
(not to scale) SEE NOTE d



BASING DIAGRAM

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
B	1.190(30.23)dia	1.310(33.27)dia
C	3.440(87.38)	3.680(93.47)
D	0.500(12.70)R	
E		0.312 (7.92)dia
F		0.375 (9.53)
G	Base: E 11-22(see notes b and c)	
Quality conformance inspection, part 2		
A		4.060(103.12)

NOTES:
a. Useful screen radius.
b. JEDEC designation.
c. Base pin alinement shall be gaged by means of a flat-plate gage 1/4 thick, with 12 holes 0.0520±0.0005 diameter and on a circle 0.6870±0.0005 diameter and 1 hole 0.3750 diameter concentric with pin circle. Chordal distance between any two adjacent pin holes shall be 0.1778±0.0005. Gaging procedure 1, as specified in JEDEC publication JO-G2-2, shall apply.
d. Orientation of deflection plates is shown as viewed from underside of base.

FIGURE 1. Outline drawing of electron tube type 1EP1.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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